Class 12 Mathematics

Chapter 10: Vector Algebra

School Worksheet: step-wise solutions

20 questions **47** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The unit vector perpendicular to the vectors $\hat{i}-\hat{j}$ and $\hat{i}+\hat{j}$ is:

(A) $\hat{k}$

(B) $-\hat{k}+\hat{j}$

(C) $\dfrac{\hat{i}-\hat{j}}{\sqrt{2}}$

(D) $\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$

Formula / idea: $\vec{a}\times\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$; unit vector $=\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

1 Take the cross product of the two vectors.

$$(\hat{i}-\hat{j})\times(\hat{i}+\hat{j})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&0\\1&1&0\end{vmatrix}$$

2 Expand along the first row.

$$=\hat{i}(0-0)-\hat{j}(0-0)+\hat{k}(1+1)=2\hat{k}$$

3 Divide by its magnitude 2 to get the unit vector. ✓ 1

$$\frac{2\hat{k}}{2}=\hat{k}$$

4 Hence the answer is option (A).

$-\hat{k}$ also works, but it is not an option.

Answer **(A)** $\hat{k}$

Watch out: Picking (D): $\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$ is one of the given directions, not perpendicular to it.

2 MCQ 1 mark

If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, $|\vec{a}|=\sqrt{37}$, $|\vec{b}|=3$ and $|\vec{c}|=4$, then the angle between $\vec{b}$ and $\vec{c}$ is:

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Formula / idea: $|\vec{b}+\vec{c}|^2=|\vec{b}|^2+|\vec{c}|^2+2\,\vec{b}\cdot\vec{c}$ and $\vec{b}\cdot\vec{c}=|\vec{b}||\vec{c}|\cos\theta$.

1 Shift $\vec{a}$ to the other side.

$\vec{b}+\vec{c}=-\vec{a}$

2 Square the magnitudes of both sides.

$|\vec{b}|^2+|\vec{c}|^2+2\,\vec{b}\cdot\vec{c}=|\vec{a}|^2$

3 Put the given values.

$9+16+2\,\vec{b}\cdot\vec{c}=37$

$\vec{b}\cdot\vec{c}=6$

4 Use $\vec{b}\cdot\vec{c}=|\vec{b}||\vec{c}|\cos\theta$. ✓ 1

$\cos\theta=\frac{6}{3\times4}=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$

5 Hence the answer is option (C).

Answer **(C)** $\frac{\pi}{3}$

Watch out: Forgetting the factor 2 in $2\,\vec{b}\cdot\vec{c}$, which gives $\vec{b}\cdot\vec{c}=12$ and $\theta=0$.

3 MCQ 1 mark

The value of λ for which $2\hat{i}+\lambda\hat{j}+\hat{k}$ and $\hat{i}-2\hat{j}+3\hat{k}$ are perpendicular is:

(A) $\frac{5}{2}$

(B) $-\frac{5}{2}$

(C) 2

(D) $\frac{3}{2}$

Formula / idea: Two non-zero vectors are perpendicular if and only if their dot product is 0.

1 Perpendicular vectors have dot product zero.

$(2\hat{i}+\lambda\hat{j}+\hat{k})\cdot(\hat{i}-2\hat{j}+3\hat{k})=0$

2 Multiply matching components and add.
$2(1) + \lambda(-2) + 1(3) = 0 \Rightarrow 5 - 2\lambda = 0$

3 Solve for λ. ✓ 1
$\lambda = \dfrac{5}{2}$

4 Hence the answer is option (A).

Answer **(A)** $\dfrac{5}{2}$

Watch out: Sign slip: writing $+2\lambda$ instead of -2λ and getting $\lambda = -\frac{5}{2}$, option (B).

4 MCQ 1 mark

If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = 8$, then $|\vec{a} \times \vec{b}|$ is:

(A) 2 **(B)** 4
(C) 6 **(D)** 8

Formula / idea: Lagrange's identity: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.

1 Write the identity linking dot and cross products.
$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$
Comes from $\sin^2\theta + \cos^2\theta = 1$.

2 Put the given values.
$|\vec{a} \times \vec{b}|^2 = (4)(25) - 64 = 36$

3 Take the positive square root. ✓ 1
$|\vec{a} \times \vec{b}| = 6$

4 Hence the answer is option (C).

Answer **(C)** 6

Watch out: Using $|\vec{a}| = 2$ and $|\vec{b}| = 5$ without squaring them, i.e. $10 - 64$.

5 MCQ 1 mark

The projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ is:

(A) $\dfrac{10}{\sqrt{17}}$

(B) $\dfrac{5\sqrt{6}}{3}$

(C) $\dfrac{10}{6}$

(D) $\sqrt{6}$

Formula / idea: Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

1 Find the dot product.
$\vec{a} \cdot \vec{b} = 2(1) + 3(2) + 2(1) = 10$

2 Find the magnitude of $\vec{b}$, the vector projected on.
$|\vec{b}| = \sqrt{1 + 4 + 1} = \sqrt{6}$

3 Use the projection formula and rationalise. ✓ 1
$\dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \dfrac{10}{\sqrt{6}} = \dfrac{10\sqrt{6}}{6} = \dfrac{5\sqrt{6}}{3}$

4 Hence the answer is option (B).

Answer **(B)** $\dfrac{5\sqrt{6}}{3}$

Watch out: Dividing by $|\vec{a}| = \sqrt{17}$ instead of $|\vec{b}|$ gives option (A), the projection of $\vec{b}$ on $\vec{a}$.

6 Assertion-Reason 1 mark

Assertion (A): The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ are collinear. **Reason (R):** Two vectors $\vec{a}$ and $\vec{b}$ are collinear if $\vec{b} = \lambda\vec{a}$ for some scalar λ.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\vec{a}$ and $\vec{b}$ are collinear $\iff$ $\vec{b} = \lambda\vec{a}$ for some scalar λ.

1 Check Assertion: compare $\vec{b}$ with $\vec{a}$.
$\vec{b} = 2\hat{i} + 4\hat{j} + 6\hat{k} = 2(\hat{i} + 2\hat{j} + 3\hat{k}) = 2\vec{a}$

2	So $\vec{b} = 2\vec{a}$; the vectors are collinear. **A is true.**	
3	Reason is the standard condition for collinear vectors. **R is true.**	
4	A follows directly from R with $\lambda = 2$. **R explains A.**	
5	Hence the correct option is (A).	✓ 1

Answer **(A) Both A and R are true and R is the correct explanation of A**

Watch out: Choosing (B) because R is a general rule; a general rule that is used to prove A does explain A.

7 **Short answer** 2 marks

Find the area of the parallelogram whose adjacent sides are $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

Formula / idea: Area of parallelogram with adjacent sides $\vec{a}, \vec{b} = |\vec{a} \times \vec{b}|$.

1	Write the formula for the area. $\text{Area} = \lvert\vec{a} \times \vec{b}\rvert$	✓ ½
2	Set up the cross product as a determinant. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix}$	
3	Expand along the first row. $= \hat{i}(1-1) - \hat{j}(1+1) + \hat{k}(-1-1)$ $= -2\hat{j} - 2\hat{k}$	✓ ½
4	Find its magnitude. $\lvert\vec{a} \times \vec{b}\rvert = \sqrt{0+4+4} = \sqrt{8} = 2\sqrt{2}$	✓ ½
5	State the area with units. $\text{Area} = 2\sqrt{2}$ sq units	✓ ½

Answer **Area $= 2\sqrt{2}$ sq units**

Watch out: Halving the answer: $\frac{1}{2}|\vec{a} \times \vec{b}|$ is for a triangle, not a parallelogram.

Level 2

16 marks

8 **Short answer** 2 marks

Find a vector of magnitude 21 units in the direction opposite to that of $\overrightarrow{AB}$, where A and B are the points $A(2, 1, 3)$ and $B(8, -1, 0)$ respectively.

Formula / idea: Vector of magnitude m opposite to $\vec{v} = -m\,\hat{v} = -m\dfrac{\vec{v}}{|\vec{v}|}$.

1 Find $\overrightarrow{AB}$ = position vector of B − position vector of A.
$\overrightarrow{AB} = (8-2)\hat{i} + (-1-1)\hat{j} + (0-3)\hat{k} = 6\hat{i} - 2\hat{j} - 3\hat{k}$ ✓ ½

2 Find its magnitude.
$|\overrightarrow{AB}| = \sqrt{36 + 4 + 9} = \sqrt{49} = 7$ ✓ ½

3 Unit vector opposite to $\overrightarrow{AB}$.
$-\dfrac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = -\dfrac{1}{7}(6\hat{i} - 2\hat{j} - 3\hat{k})$ ✓ ½

4 Multiply by 21 to get the required vector.
$21 \times \left(-\dfrac{1}{7}\right)(6\hat{i} - 2\hat{j} - 3\hat{k}) = -18\hat{i} + 6\hat{j} + 9\hat{k}$ ✓ ½

Answer $-18\hat{i} + 6\hat{j} + 9\hat{k}$

Watch out: Forgetting the minus sign for the opposite direction, or finding $\vec{A} - \vec{B}$ and then also negating it.

9 **Short answer** 2 marks

Find a unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

Formula / idea: Unit vector perpendicular to $\vec{a}$ and $\vec{b} = \pm\dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.

1 Set up $\vec{a} \times \vec{b}$ as a determinant.
$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix}$ ✓ ½

2 Expand along the first row.
$= \hat{i}(3-2) - \hat{j}(3-1) + \hat{k}(2-1) = \hat{i} - 2\hat{j} + \hat{k}$ ✓ ½

3 Find its magnitude.
$|\vec{a} \times \vec{b}| = \sqrt{1 + 4 + 1} = \sqrt{6}$ ✓ ½

4 Divide by the magnitude; both directions are perpendicular. ✓ ½

$$\text{Required unit vector} = \pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$$

Answer $\pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$

Watch out: Middle term of the expansion: the $\hat{j}$ term carries a minus sign, $-\hat{j}(3-1)$.

10 Short answer — 2 marks

Two friends flying kites from different locations find the strings of their kites crossing each other. The strings can be represented by the vectors $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$. Determine the angle formed between the kite strings. (Assume there is no slack in the strings.)

Formula / idea: $\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$.

1 Find the dot product. ✓ ½

$$\vec{a}\cdot\vec{b} = 3(2) + 1(-2) + 2(4) = 6 - 2 + 8 = 12$$

2 Find the magnitudes. ✓ ½

$$|\vec{a}| = \sqrt{9+1+4} = \sqrt{14}, \quad |\vec{b}| = \sqrt{4+4+16} = \sqrt{24}$$

3 Use the angle formula and simplify. ✓ ½

$$\cos\theta = \frac{12}{\sqrt{14}\sqrt{24}} = \frac{12}{\sqrt{336}} = \frac{12}{4\sqrt{21}} = \frac{3}{\sqrt{21}} = \frac{\sqrt{21}}{7}$$

4 Write the angle between the strings. ✓ ½

$$\theta = \cos^{-1}\left(\frac{\sqrt{21}}{7}\right)$$

Answer $\theta = \cos^{-1}\left(\frac{\sqrt{21}}{7}\right)$

Watch out: Simplifying $\sqrt{336}$ wrongly; $336 = 16 \times 21$, so $\sqrt{336} = 4\sqrt{21}$.

11 Short answer — 3 marks

Find the area of the triangle with vertices $A(1,1,2)$, $B(2,3,5)$ and $C(1,5,5)$.

Formula / idea: Area of $\triangle ABC = \dfrac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$.

1 Find $\overrightarrow{AB}$. ✓ ½

$$\overrightarrow{AB} = (2-1)\hat{i} + (3-1)\hat{j} + (5-2)\hat{k} = \hat{i} + 2\hat{j} + 3\hat{k}$$

2 Find $\overrightarrow{AC}$. ✓ ½

$\overrightarrow{AC} = (1-1)\hat{i} + (5-1)\hat{j} + (5-2)\hat{k} = 4\hat{j} + 3\hat{k}$

3 Set up the cross product.

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 0 & 4 & 3 \end{vmatrix}$$

4 Expand along the first row. ✓ 1

$= \hat{i}(6-12) - \hat{j}(3-0) + \hat{k}(4-0)$

$= -6\hat{i} - 3\hat{j} + 4\hat{k}$

5 Find its magnitude. ✓ ½

$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{36+9+16} = \sqrt{61}$

6 Area is half of this. ✓ ½

$\text{Area of } \triangle ABC = \frac{1}{2}\sqrt{61} \text{ sq units}$

Answer **Area** $= \frac{\sqrt{61}}{2}$ **sq units**

Watch out: Forgetting the $\frac{1}{2}$: $|\overrightarrow{AB} \times \overrightarrow{AC}|$ alone is the parallelogram's area.

12 **Short answer** 3 marks

Find the projection of the vector $(\vec{b} + \vec{c})$ on the vector $\vec{a}$, where $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{k}$.

Formula / idea: Projection of $\vec{p}$ on $\vec{a} = \frac{\vec{p} \cdot \vec{a}}{|\vec{a}|}$.

1 Add $\vec{b}$ and $\vec{c}$ component-wise. ✓ 1

$\vec{b} + \vec{c} = (1+1)\hat{i} + (3+0)\hat{j} + (1+1)\hat{k} = 2\hat{i} + 3\hat{j} + 2\hat{k}$

2 Write the projection formula. ✓ ½

$$\text{Projection of } (\vec{b} + \vec{c}) \text{ on } \vec{a} = \frac{(\vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a}|}$$

3 Find the dot product. ✓ ½

$(\vec{b} + \vec{c}) \cdot \vec{a} = 2(2) + 3(2) + 2(1) = 12$

4 Find $|\vec{a}|$. ✓ ½

$|\vec{a}| = \sqrt{4+4+1} = 3$

5 Divide to get the projection. ✓ ½

$$\text{Projection} = \frac{12}{3} = 4$$

Answer **Projection** $= 4$

Watch out: Dividing by $|\vec{b}+\vec{c}|$ instead of $|\vec{a}|$; divide by the vector you project **on**.

13 Short answer — 2 marks

If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$, find λ such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$.

Formula / idea: $\vec{p} \perp \vec{c} \iff \vec{p} \cdot \vec{c} = 0$.

1 Find $\vec{a} + \lambda\vec{b}$. ✓ ½

$$\vec{a} + \lambda\vec{b} = (2-\lambda)\hat{i} + (2+2\lambda)\hat{j} + (3+\lambda)\hat{k}$$

2 Perpendicular means dot product is zero. ✓ ½

$$(\vec{a} + \lambda\vec{b}) \cdot \vec{c} = 0$$

3 Multiply components with $\vec{c} = 3\hat{i} + \hat{j} + 0\hat{k}$. ✓ ½

$$3(2-\lambda) + 1(2+2\lambda) + 0(3+\lambda) = 0$$
$$6 - 3\lambda + 2 + 2\lambda = 0$$

4 Simplify and solve. ✓ ½

$$8 - \lambda = 0 \;\Rightarrow\; \lambda = 8$$

Answer $\lambda = 8$

Watch out: Treating $\vec{c}$ as having a $\hat{k}$ part; its $\hat{k}$-component is 0.

14 Short answer — 2 marks

Find the position vector of the point R which divides the line joining $P(2\vec{a} + \vec{b})$ and $Q(\vec{a} - 3\vec{b})$ externally in the ratio $1 : 2$.

Formula / idea: R divides PQ externally in $m : n$: $\vec{r} = \dfrac{m\vec{q} - n\vec{p}}{m - n}$.

1 Write the external division formula with $m = 1, n = 2$. ✓ ½

$$\vec{r} = \frac{m\,\vec{q} - n\,\vec{p}}{m - n}$$

2 Put $\vec{p} = 2\vec{a} + \vec{b}$ and $\vec{q} = \vec{a} - 3\vec{b}$. ✓ ½

$$\vec{r} = \frac{1(\vec{a} - 3\vec{b}) - 2(2\vec{a} + \vec{b})}{1 - 2}$$

3	Open the brackets and collect terms. $\vec{r} = \dfrac{\vec{a} - 3\vec{b} - 4\vec{a} - 2\vec{b}}{-1} = \dfrac{-3\vec{a} - 5\vec{b}}{-1}$	✓ ½
4	Simplify to get the position vector of R. $\overrightarrow{OR} = 3\vec{a} + 5\vec{b}$ Check: midpoint of R and Q is $2\vec{a} + \vec{b} = P$.	✓ ½

Answer $\overrightarrow{OR} = 3\vec{a} + 5\vec{b}$

Watch out: Using the internal formula $\dfrac{m\vec{q} + n\vec{p}}{m+n}$, or attaching m to $\vec{p}$ instead of $\vec{q}$.

Level 3

23 marks

15 **Short answer** 3 marks

If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\dfrac{\pi}{6}$, prove that $\vec{a} = \pm 2(\vec{b} \times \vec{c})$.

Formula / idea: A vector perpendicular to both $\vec{b}$ and $\vec{c}$ is parallel to $\vec{b} \times \vec{c}$; $|\vec{b} \times \vec{c}| = |\vec{b}||\vec{c}| \sin\theta$.

1	Given dot products are zero, so $\vec{a}$ is perpendicular to both. $\vec{a} \cdot \vec{b} = 0,\ \vec{a} \cdot \vec{c} = 0 \ \Rightarrow\ \vec{a} \perp \vec{b},\ \vec{a} \perp \vec{c}$	
2	$\vec{b} \times \vec{c}$ is also perpendicular to both, so $\vec{a}$ is parallel to it. $\vec{a} = \lambda(\vec{b} \times \vec{c})$ for some scalar λ $\vec{b} \times \vec{c} \neq \vec{0}$ since $\vec{b}, \vec{c}$ are not parallel.	✓ 1
3	Take magnitudes of both sides. $\lvert\vec{a}\rvert = \lvert\lambda\rvert\,\lvert\vec{b} \times \vec{c}\rvert = \lvert\lambda\rvert\,\lvert\vec{b}\rvert\lvert\vec{c}\rvert \sin\dfrac{\pi}{6}$	✓ ½
4	Put $\lvert\vec{a}\rvert = \lvert\vec{b}\rvert = \lvert\vec{c}\rvert = 1$ and $\sin\dfrac{\pi}{6} = \dfrac{1}{2}$. $1 = \lvert\lambda\rvert \times 1 \times 1 \times \dfrac{1}{2}$	✓ ½
5	Solve for λ. $\lvert\lambda\rvert = 2 \ \Rightarrow\ \lambda = \pm 2$	✓ ½
6	Substitute back to conclude. Hence $\vec{a} = \pm 2(\vec{b} \times \vec{c})$. Proved.	✓ ½

Answer $\vec{a} = \pm 2(\vec{b} \times \vec{c})$, **proved**

Watch out: Writing $\lambda = 2$ only; $\lvert\lambda\rvert = 2$ gives two values, which is why the result has $\pm$.

16 **Long answer** 5 marks

Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 15$.

Formula / idea: Any vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\lambda(\vec{a} \times \vec{b})$.

1 $\vec{d}$ is perpendicular to both $\vec{a}$ and $\vec{b}$, so it is parallel to $\vec{a} \times \vec{b}$.
$\vec{d} = \lambda(\vec{a} \times \vec{b})$ for some scalar λ ✓ 1

2 Set up $\vec{a} \times \vec{b}$ as a determinant.
$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 3 & -2 & 7 \end{vmatrix}$$ ✓ ½

3 Expand along the first row.
$$= \hat{i}(28 + 4) - \hat{j}(7 - 6) + \hat{k}(-2 - 12)$$ ✓ ½

4 Simplify.
$$\vec{a} \times \vec{b} = 32\hat{i} - \hat{j} - 14\hat{k}$$ ✓ ½

5 So $\vec{d}$ is:
$$\vec{d} = \lambda(32\hat{i} - \hat{j} - 14\hat{k}) = 32\lambda\hat{i} - \lambda\hat{j} - 14\lambda\hat{k}$$ ✓ ½

6 Use the condition $\vec{c} \cdot \vec{d} = 15$.
$$(2\hat{i} - \hat{j} + 4\hat{k}) \cdot (32\lambda\hat{i} - \lambda\hat{j} - 14\lambda\hat{k}) = 15$$
$$64\lambda + \lambda - 56\lambda = 15$$ ✓ 1

7 Solve for λ.
$$9\lambda = 15 \;\Rightarrow\; \lambda = \frac{5}{3}$$ ✓ ½

8 Substitute λ to get $\vec{d}$.
$$\vec{d} = \frac{5}{3}(32\hat{i} - \hat{j} - 14\hat{k}) = \frac{1}{3}(160\hat{i} - 5\hat{j} - 70\hat{k})$$ ✓ ½

Answer $\vec{d} = \frac{1}{3}(160\hat{i} - 5\hat{j} - 70\hat{k})$

Watch out: Stopping at $\vec{d} = \lambda(\vec{a} \times \vec{b})$; the condition $\vec{c} \cdot \vec{d} = 15$ fixes λ, so $\vec{d}$ is unique (no $\pm$).

17 **Short answer** 3 marks

Find a vector of magnitude 5 units perpendicular to each of $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$, where $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+3\hat{k}$.

Formula / idea: Vector of magnitude 5 perpendicular to $\vec{p},\vec{q} = \pm 5\dfrac{\vec{p}\times\vec{q}}{|\vec{p}\times\vec{q}|}$.

1	Find $\vec{a}+\vec{b}$. $\vec{a}+\vec{b}=2\hat{i}+3\hat{j}+4\hat{k}$	✓ ½
2	Find $\vec{a}-\vec{b}$. $\vec{a}-\vec{b}=0\hat{i}-\hat{j}-2\hat{k}$	✓ ½
3	Set up their cross product. $(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&4\\0&-1&-2\end{vmatrix}$	
4	Expand along the first row. $=\hat{i}(-6+4)-\hat{j}(-4-0)+\hat{k}(-2-0)$ $=-2\hat{i}+4\hat{j}-2\hat{k}$	✓ 1
5	Find its magnitude. $\sqrt{4+16+4}=\sqrt{24}=2\sqrt{6}$	✓ ½
6	Multiply the unit vector by 5; take both directions. $\pm 5\cdot\dfrac{-2\hat{i}+4\hat{j}-2\hat{k}}{2\sqrt{6}}=\pm\dfrac{5}{\sqrt{6}}(-\hat{i}+2\hat{j}-\hat{k})$	✓ ½

Answer $\pm\dfrac{5}{\sqrt{6}}(-\hat{i}+2\hat{j}-\hat{k})$, **i.e.** $\pm\dfrac{5\sqrt{6}}{6}(-\hat{i}+2\hat{j}-\hat{k})$

Watch out: Writing only one sign; both $\pm$ directions are perpendicular to $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$.

18 **Short answer** 3 marks

Using vectors, find the area of the triangle ABC with vertices $A(1,2,3)$, $B(2,-1,4)$ and $C(4,5,-1)$.

Formula / idea: Area of $\triangle ABC = \dfrac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$.

1	Find $\overrightarrow{AB}$. $\overrightarrow{AB}=(2-1)\hat{i}+(-1-2)\hat{j}+(4-3)\hat{k}=\hat{i}-3\hat{j}+\hat{k}$	✓ ½
2	Find $\overrightarrow{AC}$. $\overrightarrow{AC}=(4-1)\hat{i}+(5-2)\hat{j}+(-1-3)\hat{k}=3\hat{i}+3\hat{j}-4\hat{k}$	✓ ½

3 Set up the cross product.

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix}$$

4 Expand along the first row. ✓ 1

$= \hat{i}(12 - 3) - \hat{j}(-4 - 3) + \hat{k}(3 + 9)$

$= 9\hat{i} + 7\hat{j} + 12\hat{k}$

5 Find its magnitude. ✓ ½

$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{81 + 49 + 144} = \sqrt{274}$

6 Area is half of this. ✓ ½

$\text{Area of } \triangle ABC = \frac{1}{2}\sqrt{274} \text{ sq units}$

Answer **Area** $= \frac{\sqrt{274}}{2}$ **sq units**

Watch out: Sign errors with $-\hat{j}(\ldots)$: here $-\hat{j}(-4 - 3) = +7\hat{j}$.

19 Long answer — 5 marks

Show that the points with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ are the vertices of a right-angled triangle. Also find the remaining angles' cosines.

Formula / idea: Two sides are perpendicular if their dot product is 0; $\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$.

1 Name the points.

$\vec{A} = 2\hat{i} - \hat{j} + \hat{k}, \quad \vec{B} = \hat{i} - 3\hat{j} - 5\hat{k}, \quad \vec{C} = 3\hat{i} - 4\hat{j} - 4\hat{k}$

2 Find $\overrightarrow{AB} = \vec{B} - \vec{A}$. ✓ ½

$\overrightarrow{AB} = -\hat{i} - 2\hat{j} - 6\hat{k}$

3 Find $\overrightarrow{BC} = \vec{C} - \vec{B}$. ✓ ½

$\overrightarrow{BC} = 2\hat{i} - \hat{j} + \hat{k}$

4 Find $\overrightarrow{CA} = \vec{A} - \vec{C}$. ✓ ½

$\overrightarrow{CA} = -\hat{i} + 3\hat{j} + 5\hat{k}$

5 Take the dot product of $\overrightarrow{BC}$ and $\overrightarrow{CA}$. ✓ ½

$\overrightarrow{BC} \cdot \overrightarrow{CA} = 2(-1) + (-1)(3) + 1(5) = 0$

6 So $BC \perp CA$; the triangle is right-angled at C. ✓ ½

$\angle C = 90^\circ$

$\overrightarrow{BC}, \overrightarrow{CA}$ are non-zero and not parallel, so A, B, C are not collinear.

7 Find the side lengths. ✓ ½

$|\overrightarrow{AB}| = \sqrt{41}, \quad |\overrightarrow{BC}| = \sqrt{6}, \quad |\overrightarrow{CA}| = \sqrt{35}$

Check: $6 + 35 = 41$ (Pythagoras holds).

8 Angle A is between $\overrightarrow{AB}$ and $\overrightarrow{AC} = \hat{i} - 3\hat{j} - 5\hat{k}$. ✓ ½

$\overrightarrow{AB} \cdot \overrightarrow{AC} = -1 + 6 + 30 = 35$

9 Use the cosine formula for A. ✓ ½

$$\cos A = \frac{35}{\sqrt{41}\sqrt{35}} = \frac{\sqrt{35}}{\sqrt{41}} = \sqrt{\frac{35}{41}}$$

10 Angle B is between $\overrightarrow{BA} = \hat{i} + 2\hat{j} + 6\hat{k}$ and $\overrightarrow{BC}$. ✓ ½

$\overrightarrow{BA} \cdot \overrightarrow{BC} = 2 - 2 + 6 = 6$

11 Use the cosine formula for B. ✓ ½

$$\cos B = \frac{6}{\sqrt{41}\sqrt{6}} = \frac{\sqrt{6}}{\sqrt{41}} = \sqrt{\frac{6}{41}}$$

12 State the result.

Right angle at C; $\cos A = \sqrt{\frac{35}{41}}$, $\cos B = \sqrt{\frac{6}{41}}$

Answer **Right angle at C; $\cos A = \sqrt{\frac{35}{41}}$, $\cos B = \sqrt{\frac{6}{41}}$**

Watch out: For angle A use $\overrightarrow{AB}$ and $\overrightarrow{AC}$ (both starting at A); using $\overrightarrow{CA}$ flips the sign of the cosine.

20 Case study — 4 marks

Case Study: A toy drone starts from the origin O. It first moves to point A with $\overrightarrow{OA} = 2\hat{i} + 3\hat{j} + \hat{k}$ and is then commanded to point B with $\overrightarrow{OB} = 4\hat{i} + \hat{j} + 3\hat{k}$ (units in metres).

(i) Find the displacement vector $\overrightarrow{AB}$. 1 mark

(ii) Find the distance travelled from A to B and the unit vector along $\overrightarrow{AB}$. 1 mark

(iii) Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. 2 marks

Formula / idea: $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$; $\cos\theta = \dfrac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|}$.

(i) — 1 mark

1 Displacement = final position vector − initial position vector. ✓ ½

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

2 Subtract component-wise. ✓ ½

$\overrightarrow{AB} = (4-2)\hat{i} + (1-3)\hat{j} + (3-1)\hat{k} = 2\hat{i} - 2\hat{j} + 2\hat{k}$

Answer (i) $\overrightarrow{AB} = 2\hat{i} - 2\hat{j} + 2\hat{k}$

(ii)

1 mark

1 Distance AB is the magnitude of $\overrightarrow{AB}$. ✓ ½

$|\overrightarrow{AB}| = \sqrt{4+4+4} = \sqrt{12} = 2\sqrt{3}\text{ m}$

2 Unit vector = vector ÷ its magnitude. ✓ ½

$$\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{2\hat{i} - 2\hat{j} + 2\hat{k}}{2\sqrt{3}} = \frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$$

Answer (ii) **Distance** $= 2\sqrt{3}$ **m; unit vector** $= \frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

(iii)

2 marks

1 Find $\overrightarrow{OA} \cdot \overrightarrow{OB}$. ✓ ½

$\overrightarrow{OA} \cdot \overrightarrow{OB} = 2(4) + 3(1) + 1(3) = 14$

2 Find the magnitudes. ✓ ½

$|\overrightarrow{OA}| = \sqrt{4+9+1} = \sqrt{14}, \quad |\overrightarrow{OB}| = \sqrt{16+1+9} = \sqrt{26}$

3 Use $\cos\theta = \dfrac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|}$. ✓ ½

$$\cos\theta = \frac{14}{\sqrt{14}\sqrt{26}} = \frac{\sqrt{14}}{\sqrt{26}} = \sqrt{\frac{7}{13}}$$

4 Write the angle. ✓ ½

$\theta = \cos^{-1}\sqrt{\dfrac{7}{13}}$

Answer (iii) $\theta = \cos^{-1}\sqrt{\dfrac{7}{13}}$

Watch out: Writing $\overrightarrow{AB} = \overrightarrow{OA} - \overrightarrow{OB}$; always take head minus tail, $\overrightarrow{OB} - \overrightarrow{OA}$.